

Question_7:

Code:

```
1 // n cities start disconnected. Add one road each day and print the number
2 // of connected components and the size of the largest connected component.
3 #include <iostream>
4 using namespace std;
5
6 #define MAXN 100005
7
8 int parent_node[MAXN];
9 int sz[MAXN];
10 int num_components;
11 int max_size;
12 void initDSU(int n)
13 {
14     num_components = n;
15     max_size = 1;
16
17     for (int i = 1; i <= n; i++)
18     {
19         parent_node[i] = i;
20         sz[i] = 1;
21     }
22 }
23 int find(int i)
24 {
25     if (parent_node[i] == i)
26         return i;
27     return parent_node[i] = find(parent_node[i]);
28 }
29 int join(int i, int j)
30 {
31     int root_i = find(i);
32     int root_j = find(j);
33
34     if (root_i != root_j)
35     {
36         if (sz[root_i] < sz[root_j])
37         {
38             int temp = root_i;
```

```
39         root_i = root_j;
40         root_j = temp;
41     }
42     parent_node[root_j] = root_i;
43     sz[root_i] += sz[root_j];
44     if (sz[root_i] > max_size)
45         max_size = sz[root_i];
46
47     num_components--;
48     return 1;
49 }
50 return 0;
51 }
52 int main()
53 {
54     int n, m;
55     if (cin >> n >> m)
56     {
57         initDSU(n);
58
59         for (int i = 0; i < m; i++)
60         {
61             int u, v;
62             cin >> u >> v;
63             join(u, v);
64             cout << num_components << " " << max_size << "\n";
65         }
66     }
67     return 0;
68 }
```

Output:

Test Case_1:

```
5 6
1 2
4 2
2 3
3 3
3 1
3 3
3 4
2 4
4 5
1 5
5 4
1 5
PS C:\Users\Prabin Yadav
```

Test Case_2:

```
5 5
1 3
4 2
2 4
3 2
3 1
3 2
3 2
2 4
5 4
1 5
PS C:\Users\Prabin Yadav
```